MIT 18.06 SC - 3.3

Problem 27.1

Let A and B be any Symmetric positive definite matrices, and let λ be any eigenvalue of AB. Then we have:

$$ABx = \lambda x$$
$$(Bx)^H ABx = \lambda (Bx)^H x$$

By definition of positive definiteness of A, we get and the fact that $Bx \neq 0$ because $x \neq 0$ and B has only the zero vector in its nullspace, we get:-

$$(Bx)^H ABx > 0$$
$$\lambda (Bx)^H x > 0$$

$$\lambda (Bx)^H x = \lambda x^H B^H x$$
$$= \lambda x^H B x$$

And since $x \neq 0$ and B is positive definite :-

$$\therefore x^H B x > 0$$

Thus $\lambda > 0$

Problem 27.2

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}^T \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 1 & 5 \\ 7 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

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$$= \begin{bmatrix} x + 7y & 5x + 9y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$= x^2 + 7xy + 5xy + 9y^2$$

$$= x^2 + 12xy + 9y^2$$

$$= (x + 6y)^2 - 27y^2$$

This function is sometimes positive (e.g. when $x = y = 1$) and sometimes negative (e.g. when $x = 1$ and $y = -1$)